GROUP 6 • LLM PERFORMANCE EVALUATION

METEOR SCORE

How to evaluate AI-generated text

Word matching • Synonyms • Word order

Group 6 | METEOR Presentation

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What is METEOR?

A score that checks how close AI text is to a good reference answer.

R **Reference answer**

A sentence written by a human.

A **AI answer**

A sentence generated by a language model.

METEOR compares the two texts

Exact words | Word forms | Synonyms

Higher score = closer match to the reference

Why is METEOR useful?

AI can give different words with a similar meaning.

REFERENCE

The student is happy.

student ≈ pupil

AI ANSWER

The pupil is glad.

happy ≈ glad

1. More flexible

Can find some words with a similar meaning.

2. Better checks

Uses recall, precision, and word order.

3. Many uses

Helps check translation and summaries.

How does METEOR work?

Five easy steps to calculate the score.

1	**Read two texts**	Reference + AI answer
2	**Match words**	Exact words, stems, synonyms
3	**Calculate P and R**	Count matching words
4	**Check word order**	Find matching word groups
5	**Give final score**	Combine score and penalty

METEOR = matching quality – word order penalty

02 — THEORY

METEOR 3-Stage Word Alignment

Reference: "The student is happy." | AI Answer: "The pupil is glad."

STAGE 01

Exact Match

Identical words locked first:

"The" = "The", "is" = "is"

STAGE 02

Stem Match

Shared roots checked next:

0 matches in example

STAGE 03

Synonym Match

WordNet pairs matched last:

student ≈ pupil, happy ≈ glad

4 / 4 words matched (2 Exact + 0 Stem + 2 Synonym) — locked per stage to prevent double counting.

METEOR formula

Four small parts make one final score.

PRECISION

$$P = m / \text{candidate words}$$

RECALL

$$R = m / \text{reference words}$$

MATCH SCORE

$$F = 10PR / (R + 9P)$$

ORDER PENALTY

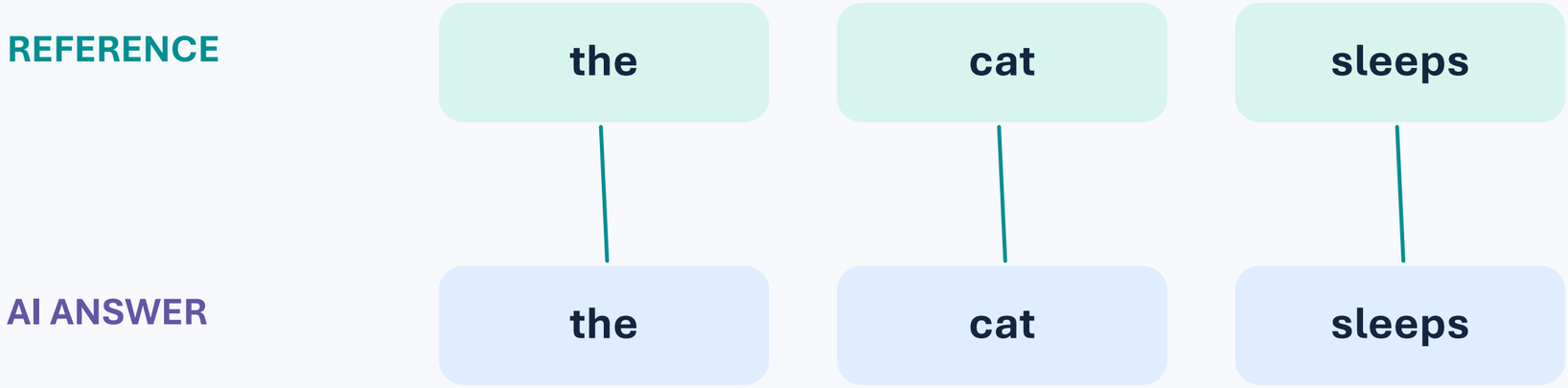
$$0.5 \times (\text{chunks} / m)^3$$

$$\text{METEOR} = F \times (1 - \text{penalty})$$

m = matched words • chunks = matched word groups

Example 1: Same words

When the words and their order are the same, the score is high.



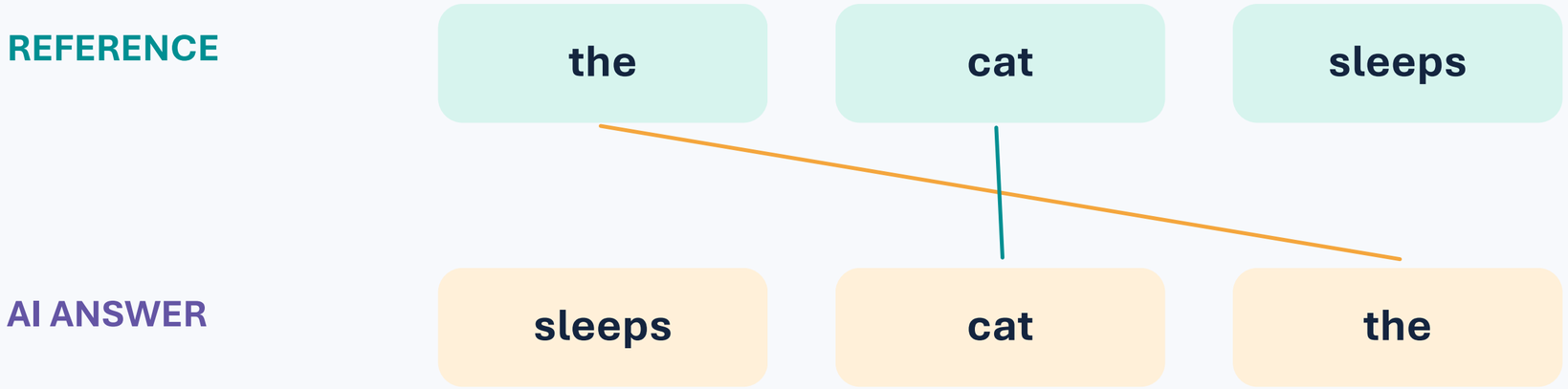
Precision = 1 Recall = 1 Chunks = 1

METEOR **0.9815**

A short exact match gets a score close to 1 (not exactly 1).

Example 2: Different word order

The same words can get a lower score when the order changes.



Precision = 1 Recall = 1 Chunks = 3

METEOR **0.5000**

More separate word groups = a bigger penalty.

Try METEOR in Python

Use NLTK: a free Python library for text processing.

meteor_demo.py

```
import nltk
from nltk.translate.meteor_score import meteor_score
nltk.download("wordnet")

ref = "the cat sleeps".split()
a = "the cat sleeps".split()
b = "sleeps cat the".split()

print(round(meteor_score([ref], a), 4))
print(round(meteor_score([ref], b), 4))
```

Install first: pip install nltk

OUTPUT

0.9815

0.5000

What to see

Same words > reversed words

Compare two model answers

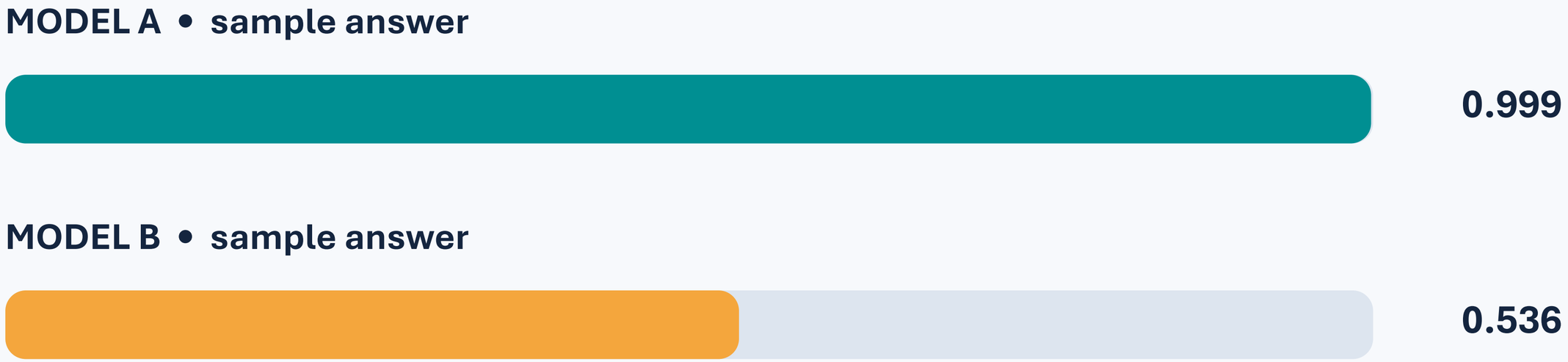
Same question. Same reference. Compare two sample answers.

QUESTION	**What is machine learning?**
REFERENCE	**Machine learning helps computers learn from data.**
MODEL A	**Machine learning helps computers learn from data.**
MODEL B	**Computers learn from data with machine learning.**

Teaching demo: these are example answers, not verified live LLM outputs.

Results: Which answer is closer?

METEOR gives higher scores to answers closer to the reference.



Model A scores higher because its words and order match the reference.

Illustrative comparison; replace with real outputs from two LLMs for a live study.

Advantages and limitations

METEOR is helpful, but it does not check everything.

+ Advantages

- Understands some synonyms
- Recognizes different word forms
- Checks precision and recall
- Easy to use with Python

− Limitations

- Needs a reference answer
- Cannot prove the answer is true
- May miss creative paraphrases
- Works differently across languages

Best practice: use METEOR with other metrics and human review.

METEOR: the key idea

Use it when you have a good reference answer to compare with.

WHAT

Compares AI text with a reference.

WHY

Checks words, forms, meaning clues, and order.

WHEN

Translation, short answers, and summaries.

METEOR is useful, but human judgment still matters.

Sources: Banerjee & Lavie (2005) • NLTK METEOR documentation

Thank you!